MILITARY SPECIFICATION

CARTRIDGE, 105MM, CS, TACTICAL, XM629 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of tactical cartridge designated as XM629 for use with 105MM Howitzers, M2A2 and M137.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.
MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.
MIL-P-46565 — Primer, Percussion, M28A2 and M28-B2, Loading, Assembling and Packing.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 — Quality Assurance Terms and Definitions.
MIL-STD-1168 — Lot Numbering of Ammunition.
MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

7549072 — Box, Packing, Ammunition, for 105 Howitzer Ammunition in Fiber Container, M105A2.
7549073 — Container, Ammo., Fiber, M105A2, for 105MM Howitzer Ammunition.
9220225 — Cartridge, 105MM, CS, Tactical, XM629.

PUBLICATIONS

ARMY

ET-9220225 — Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Material and parts shall be in accordance with applicable drawings and specifications.

3.2 **Cartridges.** Cartridges shall comply with all requirements specified on Drawing (dwg.) 9220225 and with all requirements specified in applicable specifications.

FSC 1315

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3.3 Proving ground. The projectile shall comply with the following requirements when tested as specified in 4.3.1.

3.3.1 *Functioning.* The projectile shall function satisfactorily (see 6.5) and shall burn for a minimum of 30 seconds after ejection.

3.3.2 *Premature bursts and security of parts.* There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 100 yards of flight.

3.4 Workmanship. All manufacturing, processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.
(b) Number of units of product inspected.
(c) Results obtained, by defect code, for all inspections performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verfication.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to

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an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

(a) Metal parts from one lot interfix number from one manufacturer.

(b) Cartridge cases of one lot interfix number from one manufacturer.

(c) Ejection charges of one lot interfix number from one manufacturer.

(d) Smoke assemblies loaded at one plant using the same formula and process, under the same drawing and revision and the same specification and revision.

(e) Propelling charges from not more than one lot.

(f) Loaded primers of one lot interfix number from one manufacturer.

(g) Loaded fuzes of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile, prior to assembly of expelling charge (see dwg. 9220224, covering a detail of dkg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Foreign matter in cavity	Visual	01001
Minor:	None defined.		

4.2.2.2 Projectile, prior to assembly of base plug (see dwg. 9220224 covering a detail of 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled	Visual	02001
Minor:	None defined.		

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4.2.2.3 Projectile, prior to assembly of fuze (see dwg. 9220224 covering a detail of dwg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Gap between base plug and body excessive	Gage	03001
Minor:	AQL 1.5 percent		
201.	Shear pin above flush	Visual	03002
202.	Pins loose	Visual/Manual	03003
203.	Marking missing, misleading or unidentifiable	Visual	03004
204.	Evidence of poor workmanship (see 3.4)	Visual	03005

4.2.2.4 Primer, prior to assembly to case (see dwg. 8838130 or 8838129, covering a detail of dwg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	One or more flash holes missing	Visual	04001
Major:	None defined.		
Minor:	None defined.		

4.2.2.5 Cartridge case, prior to assembling propelling charge (see dwg. 8595386 or 7548025 covering a detail of dwg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Primer missing or damaged	Visual	05001
Minor:	AQL 1.0 percent		
201.	Foreign matter in interior of case	Visual	05002
202.	Evidence of poor workmanship (see 3.4)	Visual	05003

4.2.2.6 Propelling charge, prior to assembly to Cartridge case (see dwg. 9205472 covering a detail of dwg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Increment missing	Visual	06001
102.	Any opening in bags large enough to permit escape of propellant	Visual	06002
Minor:	AQL 1.0 percent		
201.	Lacing cord broken	Visual	06003
202.	Evidence of poor workmanship (see 3.4)	Visual	06004

4.2.2.7 Cartridge case with propelling charge, prior to assembling to projectile (see dwg. 8595386 or 7548025 covering a detail of dwg. 9220225).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Incorrect sequence of increment	Visual	07001
102.	Foil side of bag, charge 5, not facing primer	Visual	07002
Minor:	None defined.		

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4.2.2.8 Cartridge assembly (see dwg. 9220225).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Primer above flush	Gage	08001
2.	Fuze not set safe (s)	Visual	08010
Major:	AQL 0.65 percent		
101.	Torque test failure of fuze	Test	08002
102.	Total length maximum	Gage	08003
103.	Any component missing	Visual	08004
104.	Staking of fuze missing	Visual	08005
Minor:	AQL 1.5 percent		
201.	Exterior coating of shell with an aggregate of bare spot in excess of ¼ square inch	Visual	08006
202.	Marking misleading, unidentifiable or missing, including incorrect color	Visual	08007
203.	Plug or fuze not fully seated	Visual	08008
204.	Evidence of poor workmanship (see 3.4)	Visual	08009

4.2.2.9 Unsealed fiber container (see dwg. 7549073).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Packing component missing	Visual	09001
Minor:	AQL 1.5 percent		
201.	Cartridge cannot be removed by hand	Manual	09002
202.	Container cap cannot be removed by hand	Manual	09003
203.	Glue or asphalt on cartridge	Visual	09004
204.	Packing components improperly assembled	Visual	09005

4.2.2.10 Sealed fiber container (see dwg. 7549073).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container cut or damaged through all asphalt in impregnated layers	Visual/Manual	10001
102.	Metal end loose or distorted	Visual/Manual	10002
103.	Tape incomplete or badly wrinkled	Visual	10003
Minor:	AQL 2.5 percent		
201.	Tear tab length, minimum	Gage	10004
202.	Contents loose	Manual	10005
203.	Gap between cover and body container exceeding ⅛ inch	Visual	10006
204.	Cuts, scuffs or gouges in outer layers	Visual	10007
205.	Marking missing, misleading or unidentifiable	Visual	10008

4.2.2.11 Sealed, wood packing box (see dwg. 7549072).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents	Visual	11001
102.	DOD symbol missing or incorrect	Visual	11002
103.	Hardware or strapping missing, broken or loose	Visual	11003
104.	Board broken	Visual	11004

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.5 percent		
201.	Strapping or hardware improperly assembled	Visual/Manual	11005
202.	Handle missing or insecure	Visual/Manual	10006
203.	Car seal missing, unsealed or improperly positioned	Visual	11007
204.	Marking missing, misleading or unidentifiable	Visual	11008

4.2.3 *Testing.*

4.2.3.1 Check test for possible determination of Primer, Percussion, M28 (see specification MIL-P-46565)—Code No. 04002. If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Govment inspector at the facility assembling the complete cartridges (see 6.4).

4.2.3.2 Bourrelet diameter (see dwg. 9220-225)—Major defect—Code No. 03006. The projectiles shall be gaged 100 percent for maximum diameter of bourrelet. Any projectile which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.3 Fit of projectile with cartridge case (see dwg. 9220225) Major defect—Code No. 12001. The cartridge case shall be gaged 100 percent for profile and diameter of the mouth to assure a proper fit with the projectile. Any case which fails to comply with the applicable requirements shall be defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4.

Note. This inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characteristics.

4.2.3.4 Proving ground (see 3.4) The following tests shall be performed as specified in 4.3.1.

4.2.3.4.1 Functioning—Major defect—Code No. 13001.—Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement, 27 cartridges shall be randomly selected from each lot for this test. Twenty-five of the cartridges shall be fired for determination of fuze functioning and observed for ejection, ignition and burning time of the 100 canister assemblies. If 3 or more projectiles fail to show evidence of fuze functioning the lot shall be rejected. If a total of 9 or more canister assemblies fail to eject, ignite or burn for the required time, the lot shall be rejected. (In the event that one or two cartridges fail to show evidence of fuze functioning, additional cartridges shall be fired in order that a total of 100 canister assemblies be observed for ejection, ignition and burning time.)

4.2.3.4.1.1 After three consecutive lots have met the criteria of 4.2.3.4.1, 11 cartridges shall be randomly selected from each lot for this test. Ten of the cartridges shall be fired for determination of fuze functioning and observed for ejection, ignition and burning time of the 40 canister assemblies. If two or more cartridges fail to show evidence of fuze functioning, the lot shall be rejected. If a total of 5 or more canister assemblies fail to eject, ignite or burn for the required time, the lot shall be rejected. (In the event that one of the cartridges fails to show evidence of fuze functioning, an additional cartridge shall be fired in order that a total of 40 canister assemblies be observed for ejection, ignition and burning time).

4.2.3.4.2 Premature bursts critical defect Code No. 13002. If any of the sample projectile (cartridges) used in any of the tests fails to comply with the requirements for premature bursts the lot shall be rejected.

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4.2.3.4.3 Refiring. If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional rounds as required.

4.2.4 *Inspection Equipment.* Equipment Tabulation Number ET–9220225 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractors' inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Proving ground test.*

4.3.1.1 *Functioning.* This test shall be conducted at a Government proving ground in accordance with the applicable acceptance test procedure. The sample cartridges shall be fired with M67 propelling charges in a gun for which the ammunition is standard. Observation shall be made for conformance to all applicable requirements. In addition, for the first three production lots from each manufacturer (loading facility) a smear camera shall be used to observe the projectile in flight, in order to record any possible malfunctions.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The cartridges shall be preserved and packaged in accordance with dwg. 7549073.

5.2 Packing.

5.2.1 *Level A.* The containers shall be packed in accordance with dwg. 7549072.

5.3 Marking. Marking of the wood box shall be in accordance with dwg. 7549072. The fiber container shall be marked in accordance with dwg. 7549073.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall resign inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design ac-

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tion required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contract or format and/or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge and shall not constitute a basis for rejection against either contractor except

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where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Functioning. Satisfactory functioning shall be defined as functioning of the fuze, and ejection, ignition and burning of the canister for the required time without detonation. (Detonation is disintegration of the canister assembly in the air after ejection from the projectile which results in a puff of smoke or dust cloud.

6.6 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN; SMUPA–ND2 and SMUPA–DC
Dover, New Jersey

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1315–A547

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